Project Title: EcoCart – Sustainable Products E-Commerce Order Management System

Week: 7

1. Basic SQL Queries

1.1 SELECT Query

Display all products

```
SQL> SELECT *
  2  FROM Product;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
 SELLER_ID CATEGORY_ID      PRICE
---------- ---------- ----------
BRAND
--------------------------------------------------------------------------------
     STOCK
----------
       101
iPhone 15
         1          1      74999

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
 SELLER_ID CATEGORY_ID      PRICE
---------- ---------- ----------
BRAND
--------------------------------------------------------------------------------
     STOCK
----------
Apple
        25


PRODUCT_ID
```

Display selected product details

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Product_ID = 101;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
 SELLER_ID CATEGORY_ID      PRICE
---------- ----------- ----------
BRAND
--------------------------------------------------------------------------------
     STOCK
----------
       101
iPhone 15
         1           1      74999

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
 SELLER_ID CATEGORY_ID      PRICE
---------- ----------- ----------
BRAND
--------------------------------------------------------------------------------
     STOCK
----------
Apple
       25
```

1.2 WHERE Clause

The WHERE clause is used to filter records based on a condition.

Find products with price less than 1000

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Price < 1000;

no rows selected
```

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
      PRICE
----------
       101
iPhone 15
       799

       102
Samsung Galaxy S24
      64999

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
      PRICE
----------

       103
OnePlus Nord CE4
      24999
```

Find products with price greater than 100

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Price > 100;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
 SELLER_ID CATEGORY_ID      PRICE
---------- ----------- ----------
BRAND
--------------------------------------------------------------------------------
     STOCK
----------
       101
iPhone 15
         1           1        799

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
 SELLER_ID CATEGORY_ID      PRICE
---------- ----------- ----------
BRAND
--------------------------------------------------------------------------------
     STOCK
----------
Apple
        25
```

1.3 ORDER BY

The ORDER BY clause is used to sort the query results.

Products from lowest to highest price

```
SQL> SELECT *
  2  FROM Product
  3  ORDER BY Price ASC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
 SELLER_ID CATEGORY_ID      PRICE
---------- ---------- ----------
BRAND
--------------------------------------------------------------------------------
     STOCK
----------
       101
iPhone 15
         1          1        799

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
 SELLER_ID CATEGORY_ID      PRICE
---------- ---------- ----------
BRAND
--------------------------------------------------------------------------------
     STOCK
----------
Apple
        25
```

Products from highest to lowest price

```
SQL> SELECT *
  2  FROM Product
  3  ORDER BY Price DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
 SELLER_ID CATEGORY_ID      PRICE
---------- ---------- ----------
BRAND
--------------------------------------------------------------------------------
     STOCK
----------
       102
Samsung Galaxy S24
         2          1      64999

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
 SELLER_ID CATEGORY_ID      PRICE
---------- ---------- ----------
BRAND
--------------------------------------------------------------------------------
     STOCK
----------
Samsung
        30
```

1.4 DISTINCT

The DISTINCT keyword is used to display unique values.

Display unique categories

```
SQL> SELECT DISTINCT C.Category_Name
  2  FROM Product P
  3  JOIN Category C
  4  ON P.Category_ID = C.Category_ID;

CATEGORY_NAME
--------------------------------------------------------------------------------
Mobiles

SQL> SELECT DISTINCT Category_ID
  2  FROM Product;

CATEGORY_ID
-----------
          1
```

2. Search Products

2.1 Search Products Based on Price

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Price BETWEEN 1000 AND 50000
  4  ORDER BY Price ASC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
 SELLER_ID CATEGORY_ID      PRICE
---------- ----------- ----------
BRAND
--------------------------------------------------------------------------------
     STOCK
----------
       103
OnePlus Nord CE4
         3           1      24999

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
 SELLER_ID CATEGORY_ID      PRICE
---------- ----------- ----------
BRAND
--------------------------------------------------------------------------------
     STOCK
----------
OnePlus
        40
```

2.2 Search Products Based on Category

```
SQL> SELECT
  2       P.Product_ID,
  3       P.Product_Name,
  4       P.Brand,
  5       P.Price,
  6       C.Category_Name
  7  FROM Product P
  8  JOIN Category C
  9       ON P.Category_ID = C.Category_ID
 10  WHERE C.Category_Name = 'Mobiles';

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND
--------------------------------------------------------------------------------
     PRICE
----------
CATEGORY_NAME
--------------------------------------------------------------------------------
       101
iPhone 15
Apple

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND
--------------------------------------------------------------------------------
     PRICE
----------
```

3. Retrieve Customer Information Display all customers

Search for a particular customer

```
SQL> SELECT *
  2  FROM Customer;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE
---------------
ADDRESS
--------------------------------------------------------------------------------
       201 Arun
Kumar
arun@gmail.com

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE
---------------
ADDRESS
--------------------------------------------------------------------------------
9876500001
Chennai


CUSTOMER_ID FIRST_NAME
```

Search for a particular customer

```
SQL> SELECT *
  2  FROM Customer
  3  WHERE Customer_ID = 201;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE
---------------
ADDRESS
--------------------------------------------------------------------------------
        201 Arun
Kumar
arun@gmail.com

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE
---------------
ADDRESS
--------------------------------------------------------------------------------
9876500001
Chennai
```

4. Apply Filtering Conditions

Multiple conditions can be combined using AND, OR, and NOT.

4.1 AND Condition

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Price > 1000
  4  AND Stock > 10;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
 SELLER_ID CATEGORY_ID      PRICE
---------- ----------- ----------
BRAND
--------------------------------------------------------------------------------
     STOCK
----------
       101
iPhone 15
         1           1      74999

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
 SELLER_ID CATEGORY_ID      PRICE
---------- ----------- ----------
BRAND
--------------------------------------------------------------------------------
     STOCK
----------
Apple
```

4.2 OR Condition

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Price < 1000
  4  OR Stock > 20;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
 SELLER_ID CATEGORY_ID      PRICE
---------- ----------- ----------
BRAND
--------------------------------------------------------------------------------
     STOCK
----------
       101
iPhone 15
         1           1      74999

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
 SELLER_ID CATEGORY_ID      PRICE
---------- ----------- ----------
BRAND
--------------------------------------------------------------------------------
     STOCK
```

4.3 IN Operator

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Price BETWEEN 1000 AND 50000;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
 SELLER_ID CATEGORY_ID      PRICE
---------- ----------- ----------
BRAND
--------------------------------------------------------------------------------
     STOCK
----------
       103
OnePlus Nord CE4
         3           1      24999

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
 SELLER_ID CATEGORY_ID      PRICE
---------- ----------- ----------
BRAND
--------------------------------------------------------------------------------
```

4.4 BETWEEN Operator

```
SQL> SELECT Product_ID, Product_Name, Brand, Price
  2  FROM Product
  3  ORDER BY Price ASC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND
--------------------------------------------------------------------------------
     PRICE
----------
       103
OnePlus Nord CE4
OnePlus
     24999


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND
--------------------------------------------------------------------------------
     PRICE
----------
       102
```

5. Basic Business Reports

5.1 Product Price Report

```
SQL> SELECT C.Customer_ID,
  2         C.First_Name,
  3         C.Last_Name,
  4         O.Order_ID,
  5         O.Order_Date,
  6         O.Total_Amount
  7  FROM Customer C
  8  JOIN Orders O
  9  ON C.Customer_ID = O.Customer_ID
 10  ORDER BY C.Customer_ID;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT
------------
         201 Arun
Kumar                                                 501 06-OCT-26
       84999

         202 Priya
Sharma                                                502 06-OCT-26
       64999

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
```

5.2 Customer Order Report

```
SQL> SELECT P.Product_ID,
  2         P.Product_Name,
  3         OD.Order_ID,
  4         OD.Quantity,
  5         OD.Unit_Price,
  6         OD.Subtotal
  7  FROM Product P
  8  JOIN Order_Details OD
  9  ON P.Product_ID = OD.Product_ID
 10  ORDER BY P.Product_ID;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------
  ORDER_ID   QUANTITY UNIT_PRICE   SUBTOTAL
---------- ---------- ---------- ----------
       101
iPhone 15
       501          1      79999      79999

       101
iPhone 15
       501          1      79999      79999

PRODUCT_ID
----------
PRODUCT_NAME
```

5.3 Product Order Report

```
SQL> SELECT C.Customer_ID,
  2         C.First_Name,
  3         C.Last_Name,
  4         COUNT(O.Order_ID) AS Total_Orders
  5  FROM Customer C
  6  LEFT JOIN Orders O
  7  ON C.Customer_ID = O.Customer_ID
  8  GROUP BY C.Customer_ID, C.First_Name, C.Last_Name
  9  ORDER BY C.Customer_ID;

CUSTOMER_ID FIRST_NAME
----------- ----------------------------------------------
LAST_NAME                                      TOTAL_ORDERS
---------------------------------------------- ------------
       201 Arun
Kumar                                                     1

       202 Priya
Sharma                                                    1

       203 Rahul
Raj                                                       1
```

5.4 Customer-wise Total Purchase Amount

```
SQL> SELECT C.Customer_ID,
  2         C.First_Name,
  3         C.Last_Name,
  4         SUM(O.Total_Amount) AS Total_Purchase_Amount
  5  FROM Customer C
  6  JOIN Orders O
  7  ON C.Customer_ID = O.Customer_ID
  8  GROUP BY C.Customer_ID, C.First_Name, C.Last_Name
  9  ORDER BY C.Customer_ID;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          TOTAL_PURCHASE_AMOUNT
-------------------------------------------------- ---------------------
        201 Arun
Kumar                                                              84999

        202 Priya
Sharma                                                             64999

        203 Rahul
Raj                                                                45999
```